TIMES SOFTWARE IDEATHON 2026

# PayShield

## Catch every payroll error and fraud before salary leaves the company.

PayShield is the hero: one readiness check before every pay run. It reads the HR modules you already use: Employee Records, Compliance & Policy Centre and Timesheet.

Read the overview →

See the live demo →

Times Software Ideathon 2026 · Prepared by Varnika Raghav · 18 October 2026

All names and figures in screens are sample data.

ONE HERO PRODUCT, READING THREE HR MODULES

**PayShield · Trust Check**
Scores every record and the whole run, and sends each alert to the module and tab where the data lives.

**Employee Records**
Employment Check · Bank Account Check

**Compliance & Policy Centre**
Salary Check · Statutory Compliance Check

**Timesheet**
Timesheet Check

★ Fraud Network

★ PayShield Agent checks the whole run in one prompt

OVERVIEW

# PayShield on one page

PayShield is the product. It reads the HR modules the team already uses and flags problems where the data lives.

**1**

## The problem

Errors and fraud are found after payday: leavers paid, salaries sent to hackers, OT above the cap, old CPF rules, unapproved raises.

See the problems →

**2**

## PayShield Trust Check

Analyses every record and gives the run a Pay Score. It only alerts; each alert opens the HR module and tab where it is fixed. Approve unlocks at 100.

Meet PayShield →

**3**

## It reads your HR modules

Employee Records (personal, bank), Compliance & Policy Centre (salary, compliance policy) and Timesheet. Flags sit on the rows where the data lives.

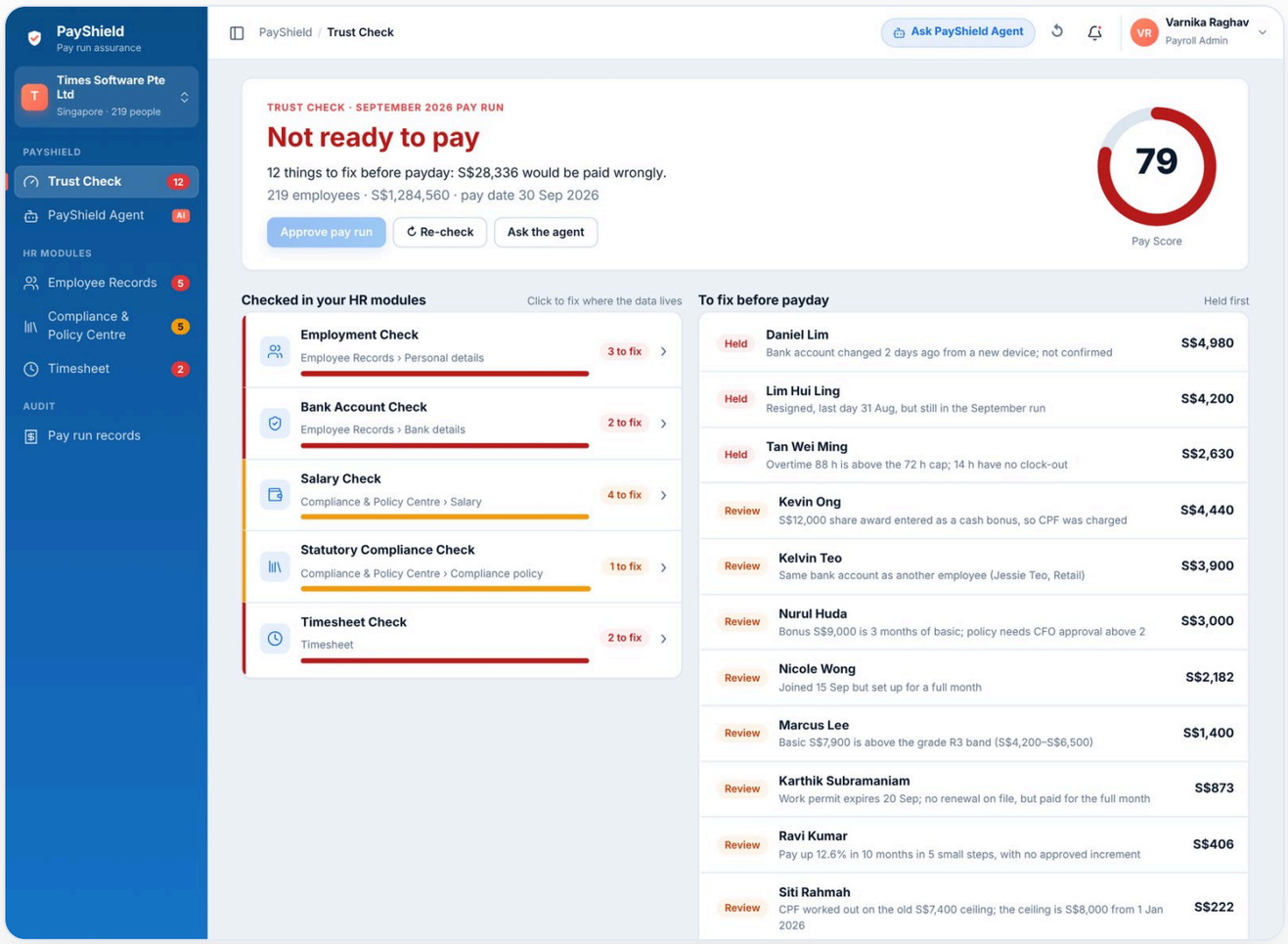
See the modules →

**4**

## PayShield Agent, AI and proof

One prompt makes the agent check the whole run and return a Pay Score with fixes; every run is recorded. Tested: 25 of 25 hidden problems caught in each of 50 synthetic companies.

See the AI →




Trust Check in the prototype: Pay Score 79, 12 alerts across 3 HR modules, S$28,336 at risk before payday

**RECORDS CHECKED**
**219**
30 checks each, before approval

**NEEDED A PERSON**
**12**
instead of reviewing all 219

**KEPT RIGHT**
**S$28,336**
in one sample run

THE PROBLEM

# Five problems our customers face every payday

| PROBLEM | WHAT HAPPENS TODAY | WHAT IT COSTS |
|---|---|---|
| **Errors found after payday** | The pay run has no check against last month or against the rules. Mistakes surface when an employee complains. | Reruns, reprocessed pay runs, recovery of overpayments, unhappy staff |
| **Salary sent to a hacker** | A stolen login can change bank details just before payday, and nothing holds that payment. | A full salary lost; the employer often pays twice |
| **Buddy punching and missed punches** | Colleagues clock in for each other; forgotten clock-outs are chased after cutoff. | Paid hours not worked; wrong OT; HR follow-ups |
| **Rule changes missed** | When MOM, CPF or IRAS changes a rule, a developer updates code and each company finds out on its own. | Short CPF, late payment interest, compliance risk |
| **HR time lost** | Payroll teams check spreadsheets by hand and answer the same pay questions every month. | Slower pay runs; less time for real HR work |

**In Times Portal today,** the pay run wizard goes from Configure Components straight to Review & Confirm. Errors are handled afterwards through *Rerun pay records* and *Reprocess pay run*. PayShield moves that work before the lock.

WHY

# Payroll mistakes are common, costly, and the rules keep moving

Singapore figures where they exist. The one US study is marked: it shows the size of the problem, not a Singapore number.

**1 in 5**

### payrolls contain errors

Missing or wrong time punches are the slowest errors to fix.

EY survey via Paycom, US, 2022

**2.84**

### salary claims per 1,000 employees in 2025

Up from 2.63 in 2024. Every claim is a pay problem an employee had to escalate to TADM.

MOM Employment Standards Report 2025

**S$8,000**

### new CPF wage ceiling from 1 Jan 2026

CPF, MOM and IRAS rules change every year. Each change needs settings updated in time.

CPF Board

### Payment-diversion fraud is rising in Singapore

Business email compromise, where attackers redirect payments to their own account, was the **3rd-largest scam type by losses** in the first half of 2026. Reported losses were **S$57.3M**, up from S$19.5M a year earlier. Payroll diversion uses the same trick on salaries: a stolen login changes the bank account. Most HR tools leave the defence as advice, not as checks inside payroll.

Singapore Police Force mid-year statistics 2026; S$ figures as reported by Scam.SG. Dayforce payroll diversion guidance.

### The market gap (checked October 2026)

- ADP, Workday and Deel check pay before payout. Singapore vendors we checked do not document it.
- None of the 8 vendors documents a hold or cooling-off on new bank details.
- None shares fraud intelligence across customers. Banks do (MAS COSMIC, Cifas, Plaid Beacon).
- None documents human-verified tracking of CPF, MOM and IRAS changes.

See the full vendor table →

WHY · MARKET CHECK

# What payroll tools offer today, checked October 2026

Eight vendors, five capabilities, from public product and help pages. ● documented · ◖ partly · — no public evidence found (which does not prove a feature is absent).

| VENDOR | CHECK BEFORE PAYOUT | BANK-CHANGE PROTECTION | ATTENDANCE FRAUD CONTROLS | STATUTORY RULE UPDATES | FRAUD INTEL SHARED ACROSS CUSTOMERS |
|---|---|---|---|---|---|
| **ADP** | ● ADP Assist flags variances | ◖ code to change; advice page | ● face-recognition kiosk | ◖ monitors rules; CPF not named | — |
| **Rippling** | ◖ time data only | ◖ alert workflow, no hold | ● GPS radius | ◖ tax updates; CPF not named | — |
| **Deel** | ● Payroll Detective | — | ◖ location check | ◖ compliance alerts | — |
| **Workday** | ● Pay Anomalies (ML) | ◖ re-authentication | ● geofence | ● auto tax updates; SG not shown | — |
| **Dayforce** | ◖ announced 2023, limited | ◖ email alert, form lock | ● fingerprint clocks | ◖ SG payroll since 2024 | — |
| **Talenox (SG)** | — | — | — | ◖ “updated with regulations” | — |
| **Info-Tech (SG)** | — | — | ● face, geofence, fake-GPS block | ◖ “regularly updated” | — |
| **QuickHR (SG)** | — | — | ◖ face; any location | — | — |
| **PayShield** | ● Trust Check on every record | ● hold, second channel, cooling-off | ● geofence, liveness, device | ● AI drafts, person verifies, what-if | ● Fraud Network, opt-in |

### No one holds a hijacked salary

The strongest controls found are alerts and re-authentication. No vendor documents a hold or cooling-off on new bank details.

### No one shares fraud intel

Banks already do it: MAS COSMIC (Singapore, 2024), Cifas (UK), Plaid Beacon (US). No payroll vendor applies the idea to salary accounts.

### No one shows CPF tracking

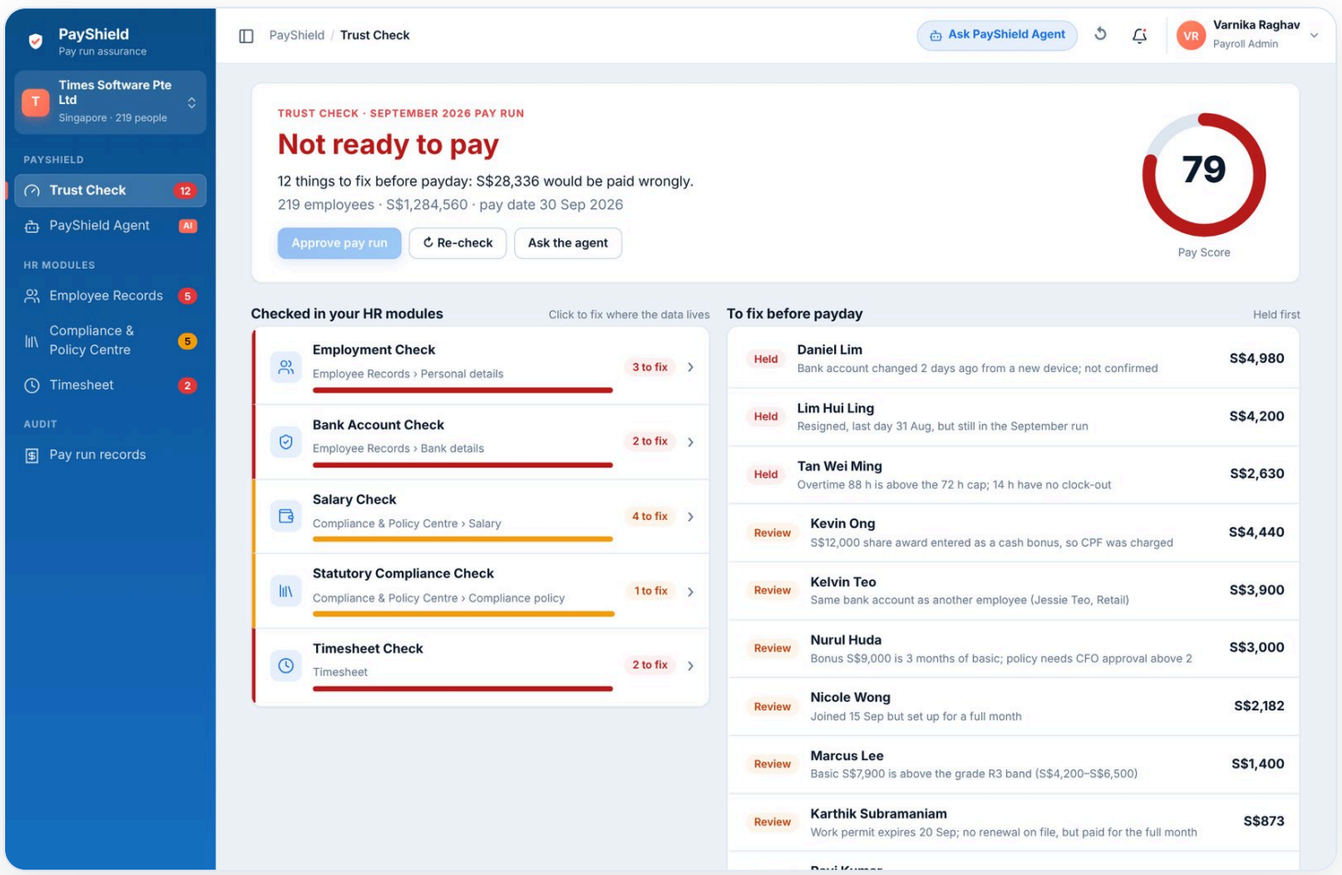
Singapore vendors say rules are “updated”. None documents automatic, human-verified tracking of CPF, MOM and IRAS changes.

Sources: adp.com and ADP press (Sep 2025, Apr 2026); rippling.com; deel.com and CPA Practice Advisor (Aug 2025); doc.workday.com; dayforce.com; talenox.com; info-tech.com.sg; quickhr.co; fenergo.com (MAS COSMIC); fca.org.uk (Cifas); plaid.com/products/beacon. Some evidence comes from customer guides or press, not vendor documentation. Confirm with each vendor before quoting externally.

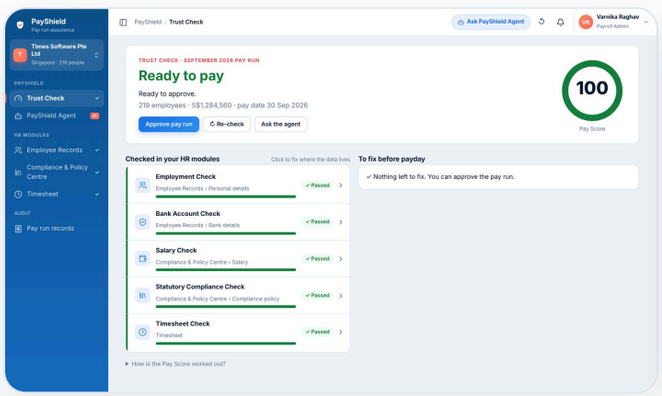
THE HERO PRODUCT

# PayShield Trust Check: analyse the run, fix where the data lives

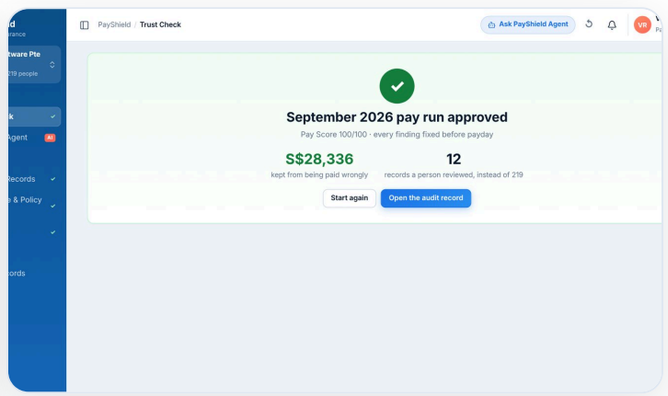
Trust Check sits between “pay run calculated” and “money sent”. It reads Employee Records, Compliance & Policy Centre and Timesheet, scores every record and the whole run, and raises alerts. It never fixes anything itself: each alert opens the module and tab where the data lives, and Trust Check re-scores the moment it is fixed.




Trust Check: Pay Score 79, with results for each HR module tab. Each row opens the module.




After fixes in the modules: Pay Score 100, Approve unlocked




Approved: S$28,336 kept from being paid wrongly

## How a pay run flows

1. **Calculate** the pay run as today
2. **Trust Check** checks all 219 records across the three HR modules and gives a Pay Score
3. **Alerts** show which module and tab failed, held first, with the amount at risk
4. **Click an alert** to open the module at that employee’s row; fix or accept it there with a reason
5. **Trust Check re-scores**; Approve unlocks at 100

## How PayShield decides

Risk = **0.40** × Salary Check + **0.35** × Bank Account Check + **0.25** × Timesheet Check, +0.03 per extra strong signal. Employment and Statutory Compliance breaches are critical rules: at least Review.

80+ Clear · 50–79 Review · under 50 Held

## What the payroll team gets

- One page that says “ready to pay” or exactly why not
- 12 records to look at instead of 219
- No new screens to learn: flags appear on the rows of the modules they already use
- Every decision and rule version recorded for audit

**A person always decides**

PayShield never changes pay by itself. It holds, explains and suggests; an admin or approver acts.

WHERE THE DATA COMES FROM

# Three HR modules, five PayShield checks

PayShield adds no new data entry. It reads the modules the payroll team already uses and puts its flags on the rows where the data lives, so every fix is made in the right place.

## Employee Records

Personal, employment and bank details

### Employment Check

on the Personal details tab

**Is every person in this run really due this pay?**

- New joiners and proration
- Leavers and final pay
- Work pass validity

**Sample run:** 3 flagged

### Bank Account Check

on the Bank details tab

**Is the salary going to the right account?**

- Bank changes near payday
- New device sign-ins
- Second-channel confirmation

**Sample run:** 2 flagged

## Compliance & Policy Centre

Salary set-up, company policies and statutory rules

### Salary Check

on the Salary tab

**Is this pay right for this person, this grade and this policy?**

- Salary vs own history
- Grade bands
- Increments and approvals

**Sample run:** 4 flagged

### Statutory Compliance Check

on the Compliance policy tab

**Are we following MOM, CPF, IRAS and our own policies?**

- Statutory rule library (versioned)
- Rule Watch: new rules drafted from official sources
- Company policies

**Sample run:** 1 flagged

## Timesheet

Clock-ins, regular hours and overtime

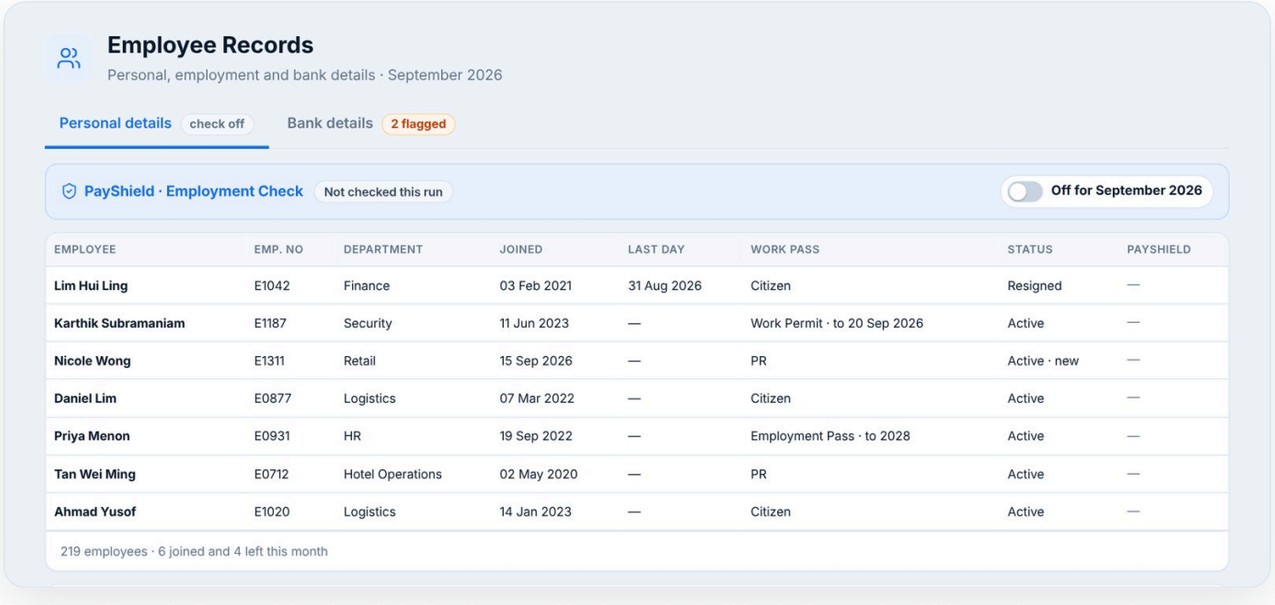
### Timesheet Check

on the Timesheet page

**Were the paid hours really worked?**

- Overtime cap and rate
- Paid hours vs clock-ins
- Buddy punching (geofence, selfie, one phone)

**Sample run:** 2 flagged




Employee Records › Personal details with the Employment Check switched off for the September 2026 run; the reason is logged

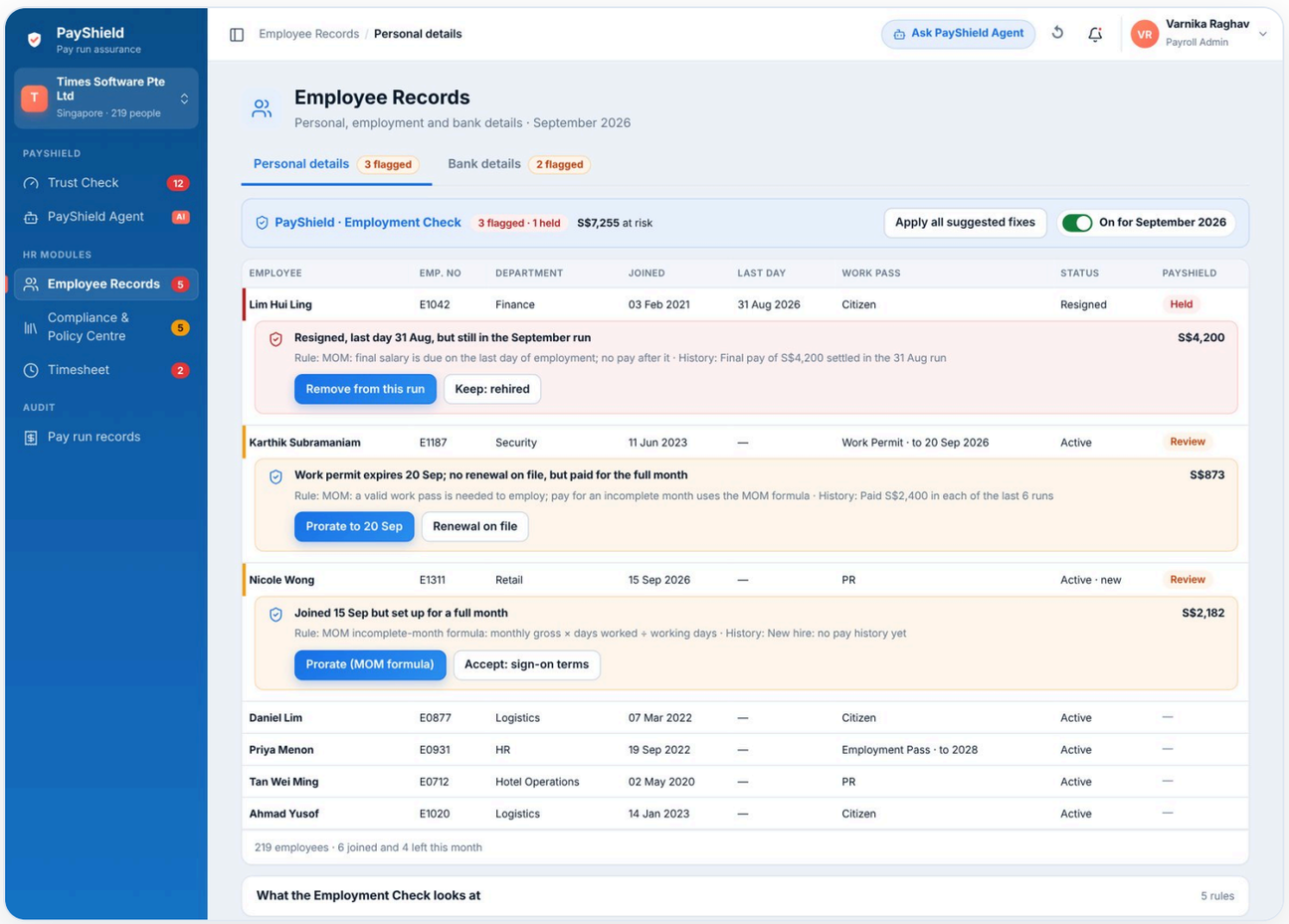
### Each check can be switched off for a pay run

- A switch on every module tab: **this pay run only** or **until turned back on**
- A reason is required; the Bank Account Check and Statutory Compliance Check also need a second person to approve
- The data stays in the module; only PayShield's check is skipped: nothing is raised, scored or asked about by the agent
- Trust Check shows the tab as "Off · not checked"; the agent reports it as skipped
- Every switch is logged in Pay run records with who, when, reason and scope

**Sold as one product.** Customers buy PayShield; checks can be switched on in stages during rollout, starting with the Bank Account Check and Statutory Compliance Check.

EMPLOYEE RECORDS › PERSONAL DETAILS · CHECK 1 OF 5

# Employment Check: is every person in this run really due this pay?




Employee Records › Personal details with PayShield flags on the rows: fix or accept where the data lives

### Found in the sample September run · 3 records · S$7,255 at risk

**Lim Hui Ling** · Resigned, last day 31 Aug, but still in the September run
Last working day: **31 Aug 2026** · rule: pay period starts 1 Sep → *Remove from this run*
Held · 52 — **S$4,200**

**Karthik Subramaniam** · Work permit expires 20 Sep; no renewal on file, but paid for the full month
Work pass valid to: **20 Sep 2026** · rule: paid to 30 Sep → *Prorate to 20 Sep*
Review · 88 — **S$873**

**Nicole Wong** · Joined 15 Sep but set up for a full month
Join date: **15 Sep 2026** · rule: 12 of 22 working days → *Prorate (MOM formula)*
Review · 66 — **S$2,182**

REAL-WORLD PROBLEM

**Leavers still paid, joiners paid a full month, work passes expiring mid-month.**

These are usually found at month-end reconciliation, after the money has gone, and recovering it is awkward for everyone.

WITH PAYSHIELD

No payments to leavers. Joiners and expiring work passes are prorated with the MOM formula before payday, not fixed after it.

### What the Employment Check looks at

| | |
|---|---|
| Last working day | **on or after the pay period start** · else Held |
| Join date in period | **prorate: gross × days worked ÷ 22 working days** · MOM formula |
| Work pass | **valid on every paid day** · else prorate to expiry |
| Final pay due | **last day · 7 days if no notice · 3 working days if dismissed** · MOM |
| Designation or pay change | **needs an approval record** · company policy |

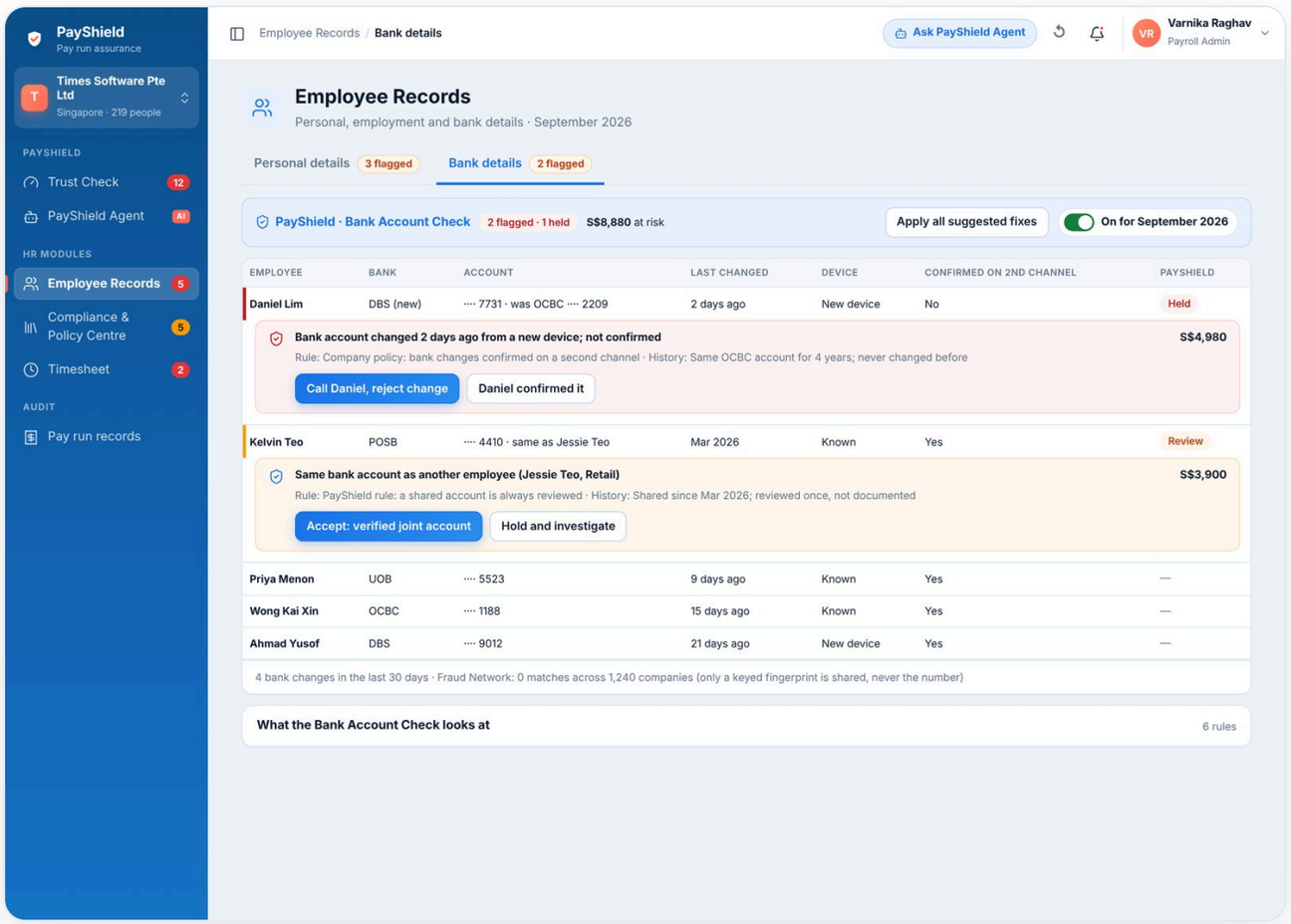
**Feeds PayShield:** A failed record check is a critical rule: the record is at least Review, and pay to a non-employee is Held.

**AI:** Explains each flag in plain English; the Agent answers “how do I prorate a joiner?” with the MOM formula.

**Try it in the demo:** Open from a Trust Check alert, or ask PayShield Agent

EMPLOYEE RECORDS › BANK DETAILS · CHECK 2 OF 5

# Bank Account Check: is the salary going to the right account?




Employee Records › Bank details with PayShield flags on the rows: fix or accept where the data lives

### Found in the sample September run · 2 records · S$8,880 at risk

**Daniel Lim** · Bank account changed 2 days ago from a new device; not confirmed
Bank risk: **+0.90** · rule: held near payday → *Call Daniel, reject change*
Held · 69 **S$4,980**

**Kelvin Teo** · Same bank account as another employee (Jessie Teo, Retail)
Shared account: **with 1 employee** · rule: always Review → *Accept: verified joint account*
Review · 66 **S$3,900**

REAL-WORLD PROBLEM

**A stolen password is enough to change a bank account a day before payday.**

The salary goes to the attacker, the employee is paid again off-cycle, and the money is rarely recovered.

WITH PAYSHIELD

A changed account is paid only after the employee confirms on a second channel. Accounts confirmed as fraud are blocked at every company on the Fraud Network.

### What the Bank Account Check looks at

| | |
|---|---|
| Changed in last 30 days | **+0.30** · bank signal |
| Within 7 days of payday | **+0.20** · bank signal |
| From a new device | **+0.20** · bank signal |
| Not confirmed on 2nd channel | **+0.20 · Held near payday** · bank signal |
| Shared with another employee | **always Review** · rule |
| Fraud Network match | **Held** · keyed fingerprint only |

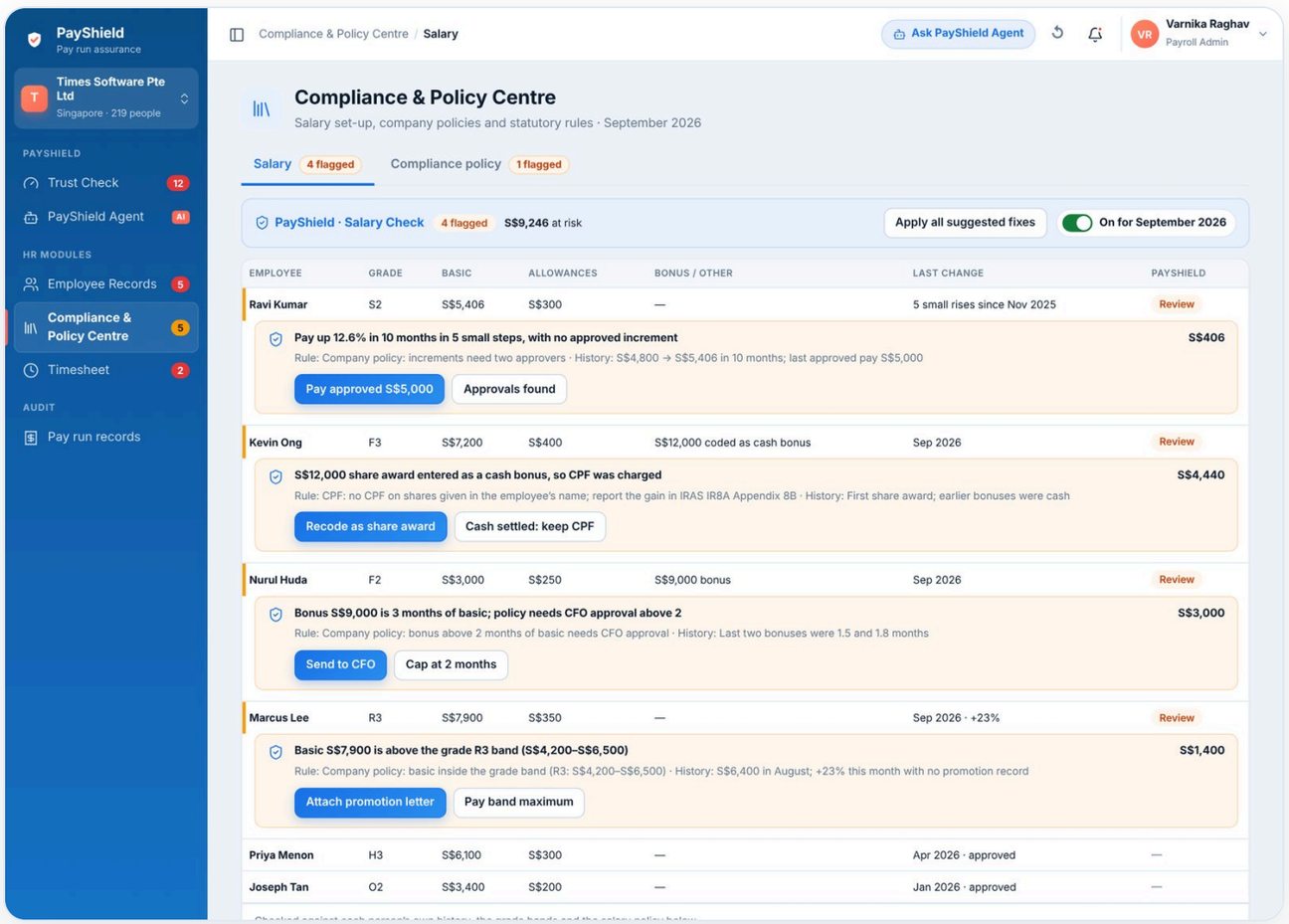
**Feeds PayShield:** Bank signal, weight 0.35. An unconfirmed new account near payday, or a Fraud Network match, is Held.

**AI:** Fraud Network shares only a keyed fingerprint of an account confirmed as fraud, never the number.

**Try it in the demo:** Open from a Trust Check alert, or ask PayShield Agent

COMPLIANCE & POLICY CENTRE › SALARY · CHECK 3 OF 5

# Salary Check: is this pay right for this person, this grade and this policy?




Compliance & Policy Centre › Salary with PayShield flags on the rows: fix or accept where the data lives

### Found in the sample September run · 4 records · S$9,246 at risk

**Ravi Kumar** · Pay up 12.6% in 10 months in 5 small steps, with no approved increment
Model v1: **0.99** · rule: flag at 0.63 → *Pay approved S$5,000*
Review · 85 **S$406**

**Kevin Ong** · S$12,000 share award entered as a cash bonus, so CPF was charged
Share award: **shares in his name** · rule: no CPF · Appendix 8B → *Recode as share award*
Review · 72 **S$4,440**

**Nurul Huda** · Bonus S$9,000 is 3 months of basic; policy needs CFO approval above 2
Bonus: **3.0 months** · rule: policy ≤ 2 months → *Send to CFO*
Review · 68 **S$3,000**

**Marcus Lee** · Basic S$7,900 is above the grade R3 band (S$4,200–S$6,500)
Grade band R3: **S$7,900** · rule: max S$6,500 → *Attach promotion letter*
Review · 76 **S$1,400**

REAL-WORLD PROBLEM

**Unapproved raises in small steps, bonuses outside policy, share awards with CPF charged, salaries outside the grade.**

Each one is small or looks normal on its own, so no single rule catches it; the cost builds up over months.

WITH PAYSHIELD

Raises, bonuses, AWS and share awards follow policy and the right CPF treatment. Slow salary creep is flagged by our own model.

### What the Salary Check looks at

| | | |
|---|---|---|
| Unusual pay | **more than 2 robust spreads (MAD) from own 6-month median** | · statistics |
| Grade band | **basic inside the band for the grade** | · company policy |
| Increment | **needs 2 approvers on record** | · company policy |
| Bonus | **≤ 2 months of basic without CFO approval** | · company policy |
| AWS | **per contract (not required by law); CPF as Additional Wage** | · MOM, CPF |
| CPF on bonus / AWS | **AW ceiling S$102,000 – OW subject to CPF** | · CPF |
| Stock awards | **shares in own name: no CPF, report in IR8A Appendix 8B; cash proceeds: CPF** | · CPF, IRAS |
| AI pattern check | **model v1 flags at ≥ 0.63** | · can only add Review |

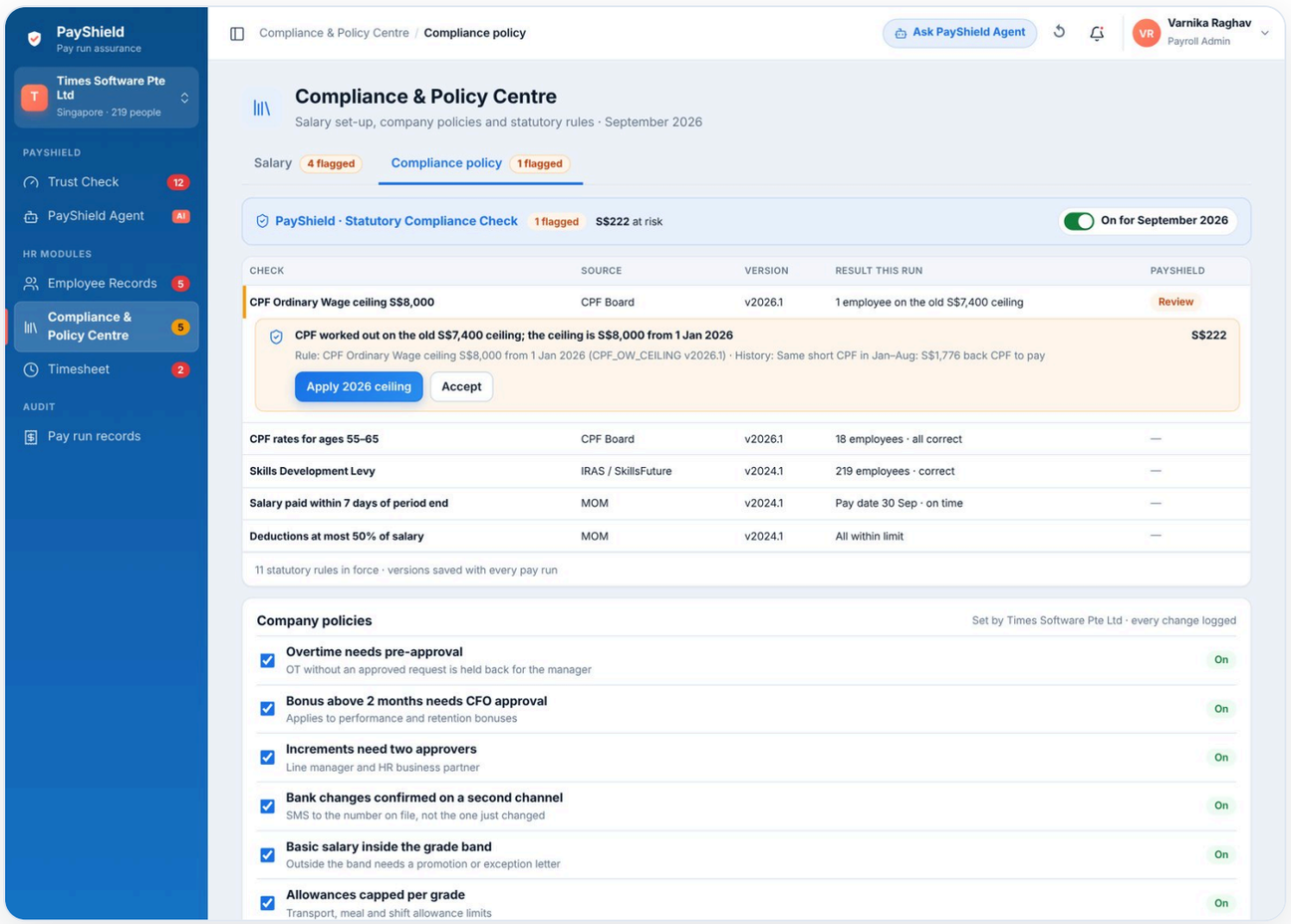
**Feeds PayShield:** Pay signal, weight 0.40. Our own model can add Review for slow patterns.

**AI:** Our own model finds salary creep and raises far above peers that no single rule catches.

**Try it in the demo:** Open from a Trust Check alert, or ask PayShield Agent

COMPLIANCE & POLICY CENTRE › COMPLIANCE POLICY · CHECK 4 OF 5

# Statutory Compliance Check: are we following MOM, CPF, IRAS and our own policies?




Compliance & Policy Centre › Compliance policy with PayShield flags on the rows: fix or accept where the data lives

### Found in the sample September run · 1 record · S$222 at risk

**Siti Rahmah** · CPF worked out on the old S$7,400 ceiling; the ceiling is S$8,000 from 1 Jan 2026 — Review · 64

CPF OW ceiling: **S$7,400 used** · rule: S$8,000 in force → *Apply 2026 ceiling* — **S$222**

REAL-WORLD PROBLEM

**CPF ceilings, rates and MOM rules change, and company policies are kept in people's heads.**

A missed setting shows up as short CPF months later, with back-payment and late-payment interest.

WITH PAYSHIELD

Every run uses today's verified statutory rules and the company's own policies, and records which versions it used for audit.

### What the Statutory Compliance Check looks at

| | |
|---|---|
| CPF Ordinary Wage ceiling | **S$8,000 / month · v2026.1** · from 1 Jan 2026 |
| CPF rates 55–65 | **new rates from 1 Jan 2026** · v2026.1 |
| SDL | **0.25% of first S$4,500 (min S$2, max S$11.25)** · IRAS / SkillsFuture |
| Salary payment | **7 days (salary) · 14 days (OT)** · MOM |
| Deductions | **≤ 50% of salary** · MOM |

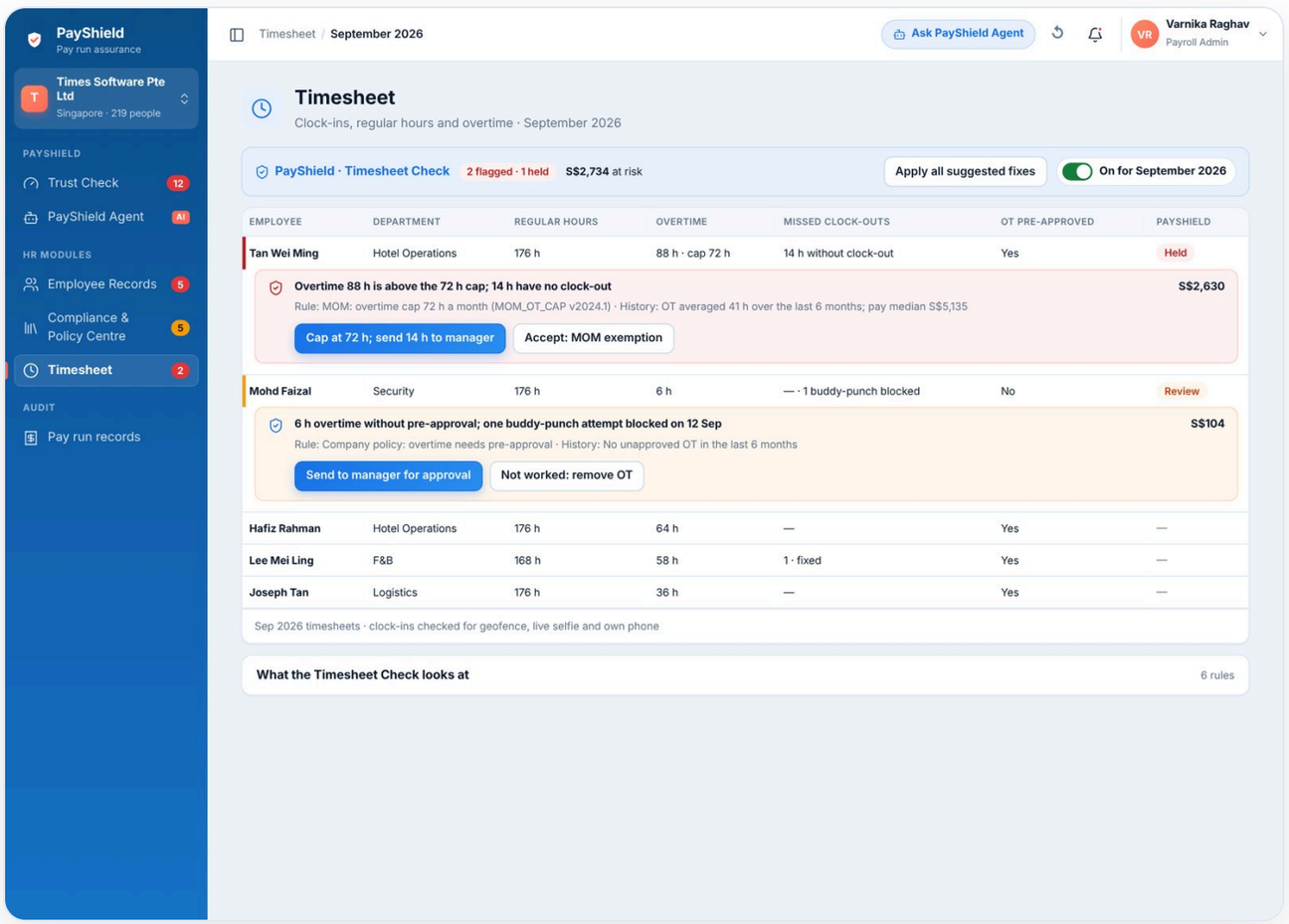
**Feeds PayShield:** A statutory rule breach is a critical rule: the record is at least Review.

**AI:** Rule Watch reads MOM, CPF Board and IRAS updates and drafts the change; a named compliance owner publishes it.

**Try it in the demo:** Open from a Trust Check alert, or ask PayShield Agent

TIMESHEET · CHECK 5 OF 5

# Timesheet Check: were the paid hours really worked?




Timesheet with PayShield flags on the rows: fix or accept where the data lives

### Found in the sample September run · 2 records · S$2,734 at risk

**Tan Wei Ming** · Overtime 88 h is above the 72 h cap; 14 h have no clock-out
Overtime: **88 h** · rule: cap 72 h → *Cap at 72 h; send 14 h to manager*
Held · 45
**S$2,630**

**Mohd Faizal** · 6 h overtime without pre-approval; one buddy-punch attempt blocked on 12 Sep
OT pre-approval: **none for 6 h** · rule: company policy → *Send to manager for approval*
Review · 77
**S$104**

REAL-WORLD PROBLEM

**Overtime above the legal cap, hours with no clock-in, and buddy punching.**

Overtime is paid on what the timesheet says. Without proof behind the hours, overpayment and MOM breaches go unseen.

WITH PAYSHIELD

Only proven hours are paid. Overtime above the cap goes to the manager, not onto the payslip, and one phone can clock in only one person.

### What the Timesheet Check looks at

| | |
|---|---|
| Overtime cap | **72 h / month** · MOM |
| OT rate | **≥ 1.5× hourly basic (12 × monthly ÷ (52 × 44))** · MOM |
| OT without clock proof | **> 0 h → Warning** · rule |
| Days without clock-out | **≥ 3 → Warning** · rule |
| Clock-in | **inside geofence + live selfie + own phone** · buddy-punch block |
| OT pre-approval | **required before the shift** · company policy |

**Feeds PayShield:** Attendance signal, weight 0.25 in the Trust Score.

**AI:** Suggests the usual clock-out time for a missed punch; the person confirms it and the manager approves.

**Try it in the demo:** Open from a Trust Check alert, or ask PayShield Agent

ONLY IN PAYSHIELD

# Two features no payroll tool we reviewed offers

Both can be demonstrated today with sample data. Neither needs real customer data for the ideathon.

★ FRAUD NETWORK

## Fraud Network: every company protects the others

**Company A**
Confirms a bank account was used in fraud

→

**Fraud Network**
Keeps only a keyed fingerprint

→

**Company B**
Same account is blocked

**Privacy**
Only a keyed one-way fingerprint (HMAC) of the account number is shared, never the number itself. No company names or employee data are shared, and customers agree to this in their contract.

**Rollout:** opt-in, starting inside multi-company groups; legal and PDPA review before any sharing between unrelated customers.

★ WHAT-IF FOR RULE WATCH

## See a rule's cost before it starts

Rule Watch reruns the company's last pay run in a sandbox with the new rule. Nothing is saved; it only shows the difference.

**Example: CPF wage ceiling S$7,400 → S$8,000 from 1 Jan 2026**
Run on Acme's November 2025 payroll, six weeks before the change (sample data)

| 46 | +S$4,210 | +S$50,520 |
|---|---|---|
| employees affected | employer CPF a month | over 12 months |




Extra employer CPF per month by pay group · totals 46 people and S$4,210

**Why it matters:** Rule Watch stops being only an alert and becomes a planning tool. Finance can budget the change, and HR can tell employees before their take-home pay moves.

**Still human-approved**
The rule behind every what-if has been verified by a compliance owner, and the company admin confirms any setting change.

WHERE THE AI IS

# Five jobs for AI, and one it deliberately does not do

AI does the work that needs judgement or language. The score that holds a salary stays plain arithmetic, so every hold can be explained to an employee, an auditor or MOM.

1

## Learns from every decision

Fewer false alarms every month

- An accepted reason (“promotion”) is remembered for 3 pay runs, then expires
- In the test run, repeat alarms fell from 6.1 to 0
- Spots patterns: “pay spike on a promotion accepted 9 of 10 times → ask for a reason code at entry”

**Guardrail:** it only suggests. An admin approves every change, and it never switches off rule or bank checks.

2

## Reads rule changes

Rule Watch

- Reads MOM, CPF Board and IRAS updates on a schedule
- Drafts the change: value, effective date, source
- What-if shows each company the cost before the rule starts

**Guardrail:** a named compliance owner verifies value, date and source before anything is published.

3

## Explains every flag

Plain English for the admin

- One sentence per flagged record: “Bank changed 2 days before payday from a new device”
- Built from a masked summary: no names, NRIC or bank numbers
- Every number checked against the record

**Guardrail:** if the AI is off or wrong, a fixed template sentence is used. Payroll never depends on AI.

## Deliberately not AI: the score

The Trust Score is rules plus statistics on each person’s own history. It is predictable, testable (50 unit tests) and explainable line by line. A black-box model should not be the thing that holds someone’s salary.

## 4 · Our own trained model

PayShield Risk Model v1 re-checks what the rules cleared and finds slow patterns such as salary creep and raises far above peers. It runs inside our system and can only add a Review flag. **See the results →**

## 5 · PayShield Agent

One prompt runs every PayShield check across the HR modules, our own model and history, and returns a Pay Score with a fix for every finding. People confirm and approve; every run is recorded. It also answers payroll questions from 41 verified answers. **See how →**

WHERE THE AI IS · OUR OWN MODEL

# Our own model finds what rules miss: 120 of 120 salary-increase cases

PayShield Risk Model v1 is a small model we train ourselves on de-identified payroll features. It looks again at records the rules cleared, and it can only add a Review flag.

**RULES ALONE**
**500 / 620**
hidden problems caught in 20 companies the model never saw

**RULES + OUR MODEL**
**620 / 620**
salary creep and raises far above peers: 0 → 120

**HARDER TEST**
**103 / 120**
smaller creep steps, smaller raises, two approvers. We report this too.

**FALSE ALARMS**
**2.4%**
of clean records, from 1.6% with rules alone

## What it learned, from 9,344 records

Top weights of 15 inputs. Each flag shows its top 3 reasons, e.g. “pay growth over 10 months not covered by approved increments”.

| | |
|---|---|
| Pay growth over 10 months not covered by approved increments | +4.22 |
| Small unexplained pay steps over 10 months | +2.74 |
| Raise compared with peers in the same grade | +2.43 |
| Same approver signed several off-cycle raises this month | +1.97 |
| Pay unusual for this person (vs own 6 months) | +0.94 |
| Raise outside the company’s increment cycle | +0.65 |
| A reason was recorded for the change | +0.63 |

## Security by design

- **Runs inside our system.** Training and scoring need no outside AI service.
- **15 allow-listed inputs.** No ids, names, NRIC, bank numbers, gender, age, race or nationality. The build fails if one is added.
- **Can only add Review.** It never clears a rule flag, holds pay or changes pay.
- **Hard to poison.** Retrained only from two-person decisions; each version needs two approvers.
- **Drift alarm.** If its flag rate leaves 0.5–3%, it switches to rules-only.

!

### Two models of our own, two jobs

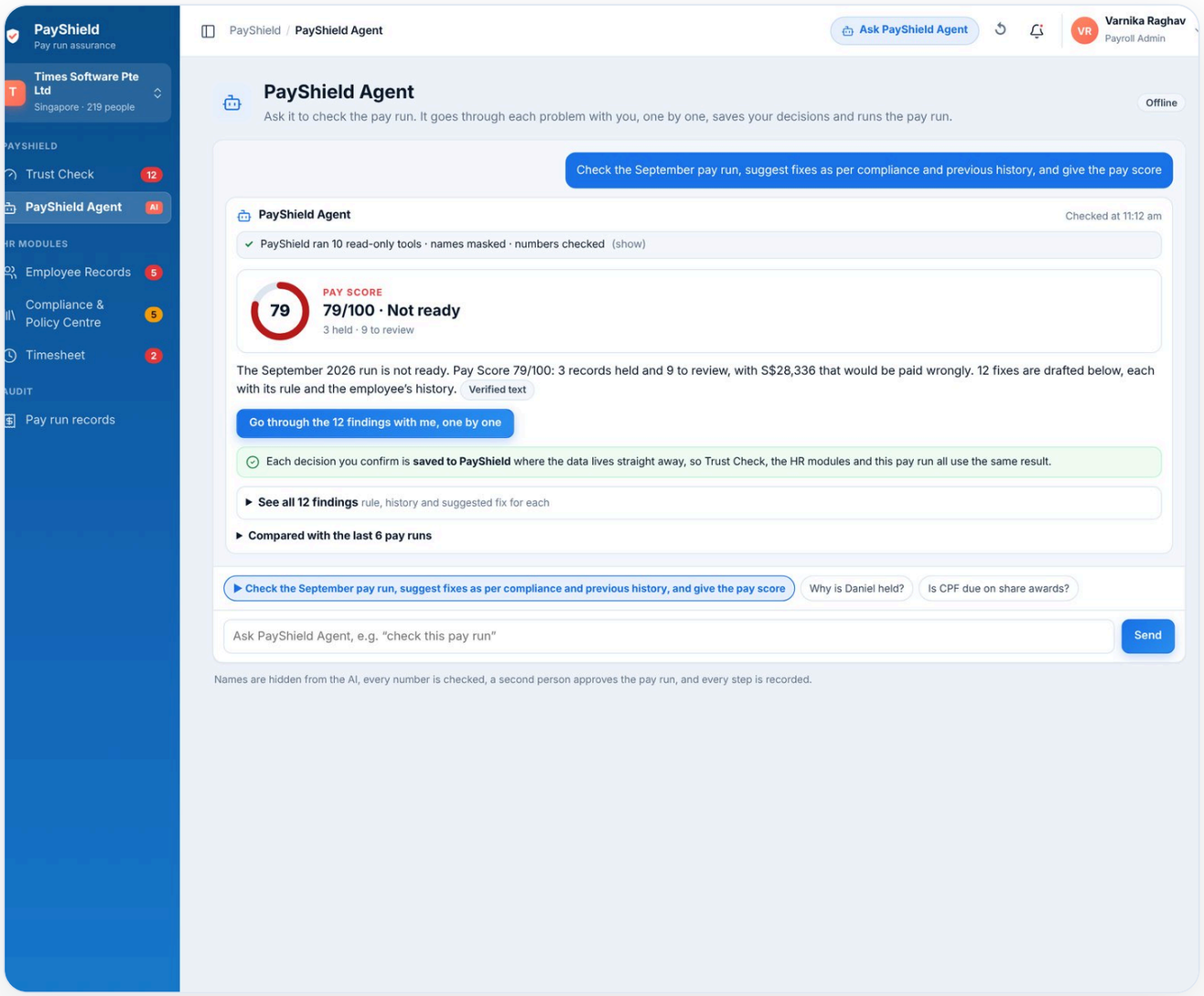
This small risk model finds the patterns and can be explained input by input. PayShield Agent, our payroll language model, answers questions and words the results, from masked data only. See the agent →

Synthetic data: we planted these patterns, so this shows the model can learn them, not how it performs on real payroll. Next: retrain on pilot customers’ audited decisions. Reproduce with `npm test` or the demo’s “Our own model” tab.

WHERE THE AI IS · PAYSHIELD AGENT

# One prompt runs the whole of PayShield

"Check the September pay run, suggest fixes as per compliance and previous history, and give the pay score." The agent runs all of PayShield in the background and answers with a Pay Score and a fix for every finding.




PayShield Agent in the prototype: 10 read-only tools, then Pay Score 79/100 and 12 fixes with their rule and history

## What runs in the background

`payrun.read`
Read the pay run

`records.bank`
Bank Account Check + Fraud Network

`model.v1.score`
Our own risk model

`timesheet`
Timesheet Check

`payscore.compute`
Pay Score

`records.personal`
Employment Check

`policy.salary`
Salary Check

`policy.compliance`
Statutory Compliance Check

`history.compare`
Last 6 pay runs

`fixes.draft`
Fixes with rule and history

All read-only. None of them can change pay.

## What comes back

- **Pay Score** for the whole run: half records ready, half pay value safe. Held records cap it at 79; open reviews at 89.
- **Every finding** with its rule (MOM, CPF, IRAS or company policy) and the employee's history
- **A suggested fix** and the amount at risk

## Then it asks, you decide, it runs

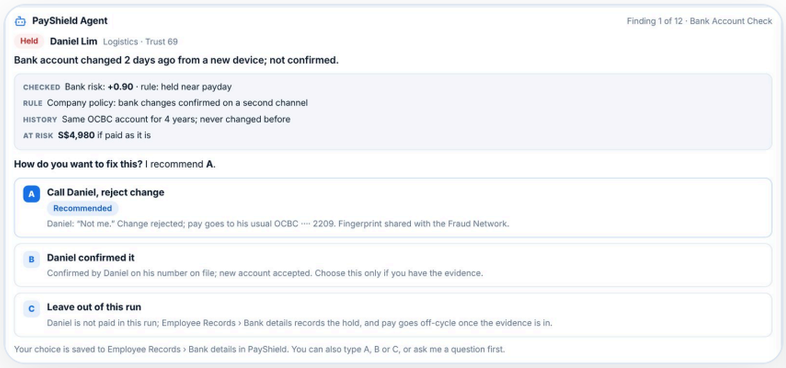
Agent checks → Asks about each finding → You choose A, B or C → Saved to its module → Approver signs off → Agent runs the pay run → Recorded

Each choice is saved to its HR module as you go, so PayShield is up to date before the pay run starts.

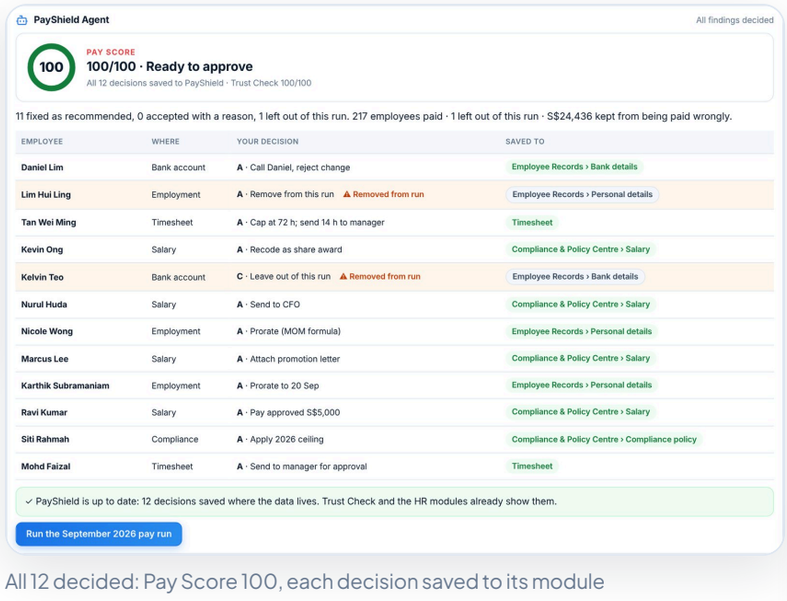
WHERE THE AI IS · PAYSHIELD AGENT

# It asks about each finding, then runs the pay run

Held findings first. For each one the agent shows what it found, the rule, the employee's history and the money at risk, recommends an option and explains why. You choose A, B or C, and the choice is saved in the HR module where the data lives. When all are decided, an approver signs off and the agent runs the pay run and records it.



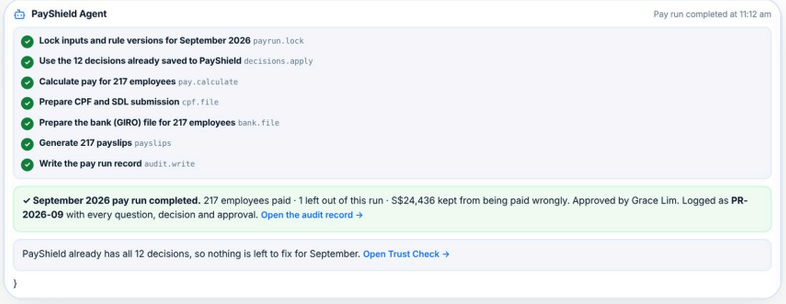

Finding 1 of 12: Daniel Lim, Bank Account Check, held. Options with what each does to this run.




All 12 decided: Pay Score 100, each decision saved to its module

## Every decision is saved to PayShield

- Each A, B or C you confirm is **saved to its HR module straight away**; Trust Check re-scores after every answer (79 → 100)
- The module row shows "Decided with PayShield Agent" with who and when, plus Undo
- No temporary fixes: Trust Check, the HR modules and the pay run always agree
- Only a decision the user confirms, one at a time, can change data; a prompt alone cannot




Final run: lock inputs, use the saved decisions, calculate, CPF and GIRO files, payslips, audit record

**Audit record PR-2026-09** keeps the prompt, every tool call, each question with the option chosen, who chose it and when, the approver and the rule versions. It also answers payroll questions: 41 verified answers, 82% right first time on unseen wording.

PROOF · TEST RUN

# We tested it: 25 of 25 hidden problems caught

A made-up company of 500 employees with 12 months of pay, normal noise (increments, OT swings, promotions, bank changes) and 25 hidden problems. The real `scoring.js` checks it. Repeated on 50 random companies.

**CAUGHT**

**25 / 25**

in every one of the 50 runs. First design: 18 on average (17–19).

**FALSE ALARMS**

**1.4%**

of clean records (worst run 2.1%). First design: 3.6%. Target under 5%.

**REPEAT ALARMS**

**0**

the month after admins accept a reason. Without learning: 6.1 per run.

**REAL DATA USED**

**None**

Seeded and repeatable: anyone can rerun it with `npm test` or in the demo.

| HIDDEN PROBLEM · ALL 50 COMPANIES | FIRST DESIGN | CURRENT |
|---|---|---|
| Bank account changed by a hacker 1–2 days before the run | 150 / 150 | **150 / 150** |
| Bank account changed by a hacker 2–3 weeks earlier | 100 / 100 | **100 / 100** |
| Hacker also controls the employee's email and confirms the alert | 5 / 50 | **50 / 50** |
| Ghost employee paid into another employee's account | 2 / 100 | **100 / 100** |
| Pay 35–90% above normal with no reason | 250 / 250 | **250 / 250** |
| Wrong CPF contribution | 200 / 200 | **200 / 200** |
| Ex-employee still in the pay run | 150 / 150 | **150 / 150** |
| Overtime paid with no clock-in proof | 20 / 150 | **150 / 150** |
| Full pay for days with no clock-out | 23 / 100 | **100 / 100** |

### What the test taught us: 4 fixes

1. **Shared account → always Review.** The ghost-employee pattern scored Clear before.
2. **OT paid without clock proof → Warning rule.**
3. **Pay for 3+ days with no clock-out → Warning rule.**
4. **Second-channel confirmation** for a new account from a new device, so a hijacked email cannot approve it. Hourly pay that matches verified hours is no longer flagged, which cut false alarms.

!

### Read this honestly

We wrote both the problems and the checks, so 25 of 25 shows coverage of the patterns we know. Next: the same test on anonymised past pay runs from 2 pilot customers.

TRUST SCORE

# How the score is calculated, and when a record is reviewed

The score is plain arithmetic from three signals. It is the same logic running in the prototype's `scoring.js`, covered by 16 unit tests.

## 1 · THREE SIGNALS, EACH 0 TO 1

### Pay variance

Highest of: rule severity (Critical 0.9, Warning 0.6, Info 0.3) and distance from the person's own 6-month median. A bonus, increment or proration cancels the distance part.

### Bank-change risk

Changed in 30 days +0.3 · within 7 days of payday +0.2 · new device +0.2 · not confirmed +0.2 · shared account +0.5 · Fraud Network match = 1.0

### Attendance integrity

Missing punches, OT without clock-in proof, blocked buddy punches, geofence exceptions.

## 2 · COMBINE

risk = 0.40 × pay + 0.35 × bank + 0.25 × attendance
+ 0.03 for each extra signal ≥ 0.50

**Trust Score = 100 × (1 − risk)**

### Worked example: Daniel (demo)

| | |
|---|---|
| Pay 0.40 × 0.82 | 0.328 |
| Bank 0.35 × 0.90 | 0.315 |
| Attendance 0.25 × 0.40 | 0.100 |
| Two strong signals | +0.030 |
| **Risk** | **0.773** |
| **Trust Score** | **23 · Held** |

## 3 · DECIDE

80–100 Clear — paid as normal

50–79 Review — admin fixes or accepts with a reason

0–49 Held — second approver releases it

### Hard overrides win over the score

Any Critical rule (wrong CPF, ex-employee still paid, OT below 1.5×) → at least Review. Fraud Network match, or a new bank account not confirmed near payday → Held.

### Tuning

Weights and thresholds are settings per company. In the pilot, tune so all planted errors are caught while only 3–5% of records are flagged; keep 70%+ of flags acted on.

SECURITY

# The AI never sees personal data

Our own rules engine does every check inside the payroll system. The AI only turns the result into one readable sentence, from a masked summary.

INSIDE TIMES PORTAL · PERSONAL DATA STAYS HERE

**Payroll data**
Pay, bank details, timesheets, contracts

→

**Rules engine**
Computes signals and the Trust Score

→

**Masking**
Token + flag codes + 2–3 numbers only

**Admin sees** Real name + checked sentence + score. The token is swapped back to the name inside Times Portal.

⇄ masked only

**AI service (Azure OpenAI)**
Writes one sentence. Prompts are not used to train models and are not shared with OpenAI.

```
{
  "subject": "EMP_1AE53A",
  "flags": ["BANK_CHANGED_RECENT",
            "NEW_DEVICE", "PAY_SPIKE"],
  "values": { "days_before_payday": 2,
              "pay_change_pct": 63 },
  "score": 23, "status": "HELD"
}
```

| CONTROL | WHAT WE DO | STANDARD |
|---|---|---|
| **Data minimisation** | Allow-listed payload only. Never names, NRIC/FIN, bank numbers, addresses or salary history. | PDPA · OWASP LLM02 |
| **Pseudonymisation** | New token per employee per pay run (HMAC-SHA256, key in Azure Key Vault). The token map never leaves our database. | ISO/IEC 27001 controls |
| **AI provider** | Azure OpenAI in an approved geography, via private endpoint. Apply for modified abuse monitoring so prompts aren't stored. | Microsoft data privacy terms |
| **Safe output** | Every number in the AI sentence must match the payload, otherwise a fixed template is used. Free-text fields never enter prompts. | OWASP LLM01 · LLM05 · LLM09 |
| **Works without AI** | Scores and holds come from the rules engine. If the AI is off, payroll checks still run. | Resilience by design |
| **Access and encryption** | Role-based access (managers see only their team), MFA for admins, TLS 1.2+ in transit, AES-256 at rest. | ISO/IEC 27001 controls |
| **Audit and governance** | Every score, flag and decision logged; prompts with personal data never logged. Data protection impact assessment before the pilot. | PDPA |

SECURITY · BEYOND THE AI

# How PayShield stays secure

Payroll data is a prime target. Besides keeping personal data away from the AI, four layers protect the data, the people who can change it, and PayShield itself.

### Data where it lives

Even if a server or backup is stolen

- AES-256 encryption at rest, with keys the company controls (Azure Key Vault)
- Extra field-level encryption for NRIC/FIN, bank account and salary
- TLS 1.2+ for everything in transit
- Screens show only the last 4 digits; seeing more needs a reason and is logged
- Each company's data kept apart by the database itself (row-level security)

### Accounts and approvals

Even if a password is stolen

- Phishing-resistant MFA for admins: passkeys or an authenticator app, not SMS
- MFA asked again for a bank change, releasing a hold or approving a pay run
- Two-person rule: nobody approves their own pay or their own change
- Least privilege: managers see only their own team
- Short sessions, rate limits and alerts on unusual sign-ins

### Monitoring and audit

Even if someone inside misuses access

- Tamper-proof, write-once log of every score, flag and decision
- Personal data never written to logs or prompts
- Alerts on bulk exports, many bank changes at once, or admin actions at odd hours
- No passwords in code: secrets in Key Vault, services use managed identities

### Build and compliance

So it stays secure release after release

- Threat model at design time; OWASP ASVS Level 2 as the build checklist
- Code and dependency scanning on every build; penetration test before go-live
- PDPA: impact assessment before the pilot, data kept only as long as needed
- Breach plan: notify the PDPC within 3 calendar days once a breach is assessed notifiable
- ISO/IEC 27001-aligned controls; SOC 2 report as a later goal

★ FRAUD NETWORK

### Fingerprints are keyed, not just hashed

A plain hash is not enough: bank account numbers are short, so an attacker could hash every possible number and reverse the list.

**DBS ···· 7731**
stays with the company

→

**HMAC-SHA256**
secret key locked in a hardware security module (HSM)

→

**9f3a…c71e**
only this is shared

Without the key, a leaked fingerprint list is useless. Later phase: **private set intersection**, so companies find matches without either side revealing its own list.

### For the ideathon: sample data only

The demo and this proposal use made-up names, amounts and accounts. No real employee or customer data has been used.

BUSINESS CASE

# A feature sales can demo, price and defend

PayShield gives sales a clear story: *"We catch payroll errors and fraud before salary is paid, and we keep you in line with Singapore rules."*

WIN NEW DEALS

### A demo that sells itself

- Show a fraud being stopped live in 5 minutes
- Singapore-specific checks global tools don't offer
- Answers the CFO's question: "How do we know payroll is right?"

GROW REVENUE

### A paid add-on

- Sold as "PayShield" through the existing Plan → Add-ons flow
- Pricing per employee per month, or bundled in a premium plan (to decide)
- What-if helps finance budget each rule change, a reason to buy for CFOs

KEEP CUSTOMERS

### Harder to leave

- Fewer payroll complaints, so fewer reasons to switch
- Audit trail and rule history build up inside Times Portal
- Fraud Network: every company protects the others

### Best-fit customers

| | |
|---|---|
| **Construction, marine, logistics, F&B, cleaning** | Shift work, heavy OT, many sites and foreign workers |
| **SMEs with a small HR team** | No in-house compliance expert to track MOM and CPF changes |
| **Multi-entity groups** | Many pay runs a month; one bad run is costly |

### What we will measure in the pilot

| METRIC | PROPOSED TARGET |
|---|---|
| Reruns after lock | –50% |
| Salaries paid to a fraudulent account | Zero |
| Demo-to-deal win rate where PayShield is shown | Track vs baseline |
| Add-on attach rate in new deals | Track from launch |

Targets are proposals to confirm with pilot customers and sales leadership.

PRICE BENCHMARK

Singapore payroll list prices per employee per month: Talenox S$5 per extra employee, QuickHR S$3–9, Employment Hero S$6–8, Info-Tech from S$2. Add-ons elsewhere cost about 20–50% of the base price (PeopleHR £1.50–2 on £3–9.50).

Vendor pricing pages, Oct 2026, excl. GST

PROPOSED, FOR SALES TO CONFIRM

**S$2.00** per employee per month, **S$1.50** above 200 employees, minimum S$60 a month. 30 customers × 200 employees × S$1.50 = **S$108,000 a year**. A 100-person company pays S$2,400 a year: less than one salary lost to a hijacked bank account.

PLAN AND ASK

# Start small, show value in the first month

PHASE 1 · ABOUT 2 SPRINTS

### Salary Check + Trust Check

~10 rules on data we already hold, plus the test run repeated on anonymised past pay runs.

PHASE 2 · ABOUT 1 SPRINT

### Rule Watch + AI explanations

Versioned rule library with reviewer approval, what-if impact, plain-English reasons on every flag.

PHASE 3 · ABOUT 2 SPRINTS

### Bank Account Check + Fraud Network

Second-channel confirmation, cooling-off, approvals card. Fraud Network opt-in, inside company groups first.

PHASE 4 · 1–2 SPRINTS

### Timesheet Check

Geofence, device binding, missed-punch nudges, liveness via an SDK.

## What we need from Heads of Departments

- Approval to build Phase 1 and 2
- A small squad: backend, frontend and QA
- A named compliance owner to verify rule updates
- Two pilot customers with shift work and OT, and anonymised past pay runs to re-test on
- Legal and PDPA review of Fraud Network sharing

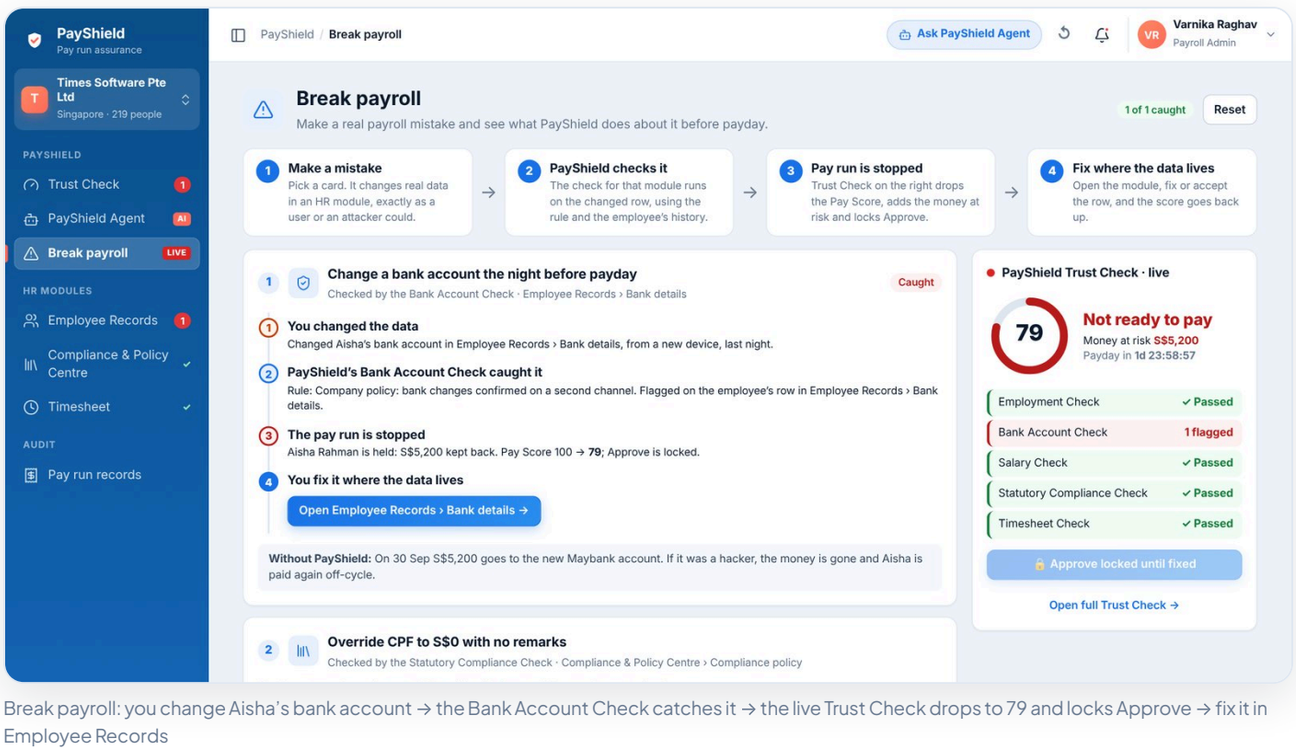
## Why Phase 1 first

It uses only data already in Times Portal, needs no consent changes, and reduces reruns from the first pay run. Sprint estimates are early assumptions for the engineering team to confirm after reviewing the backend.

LIVE DEMO

# See PayShield working end to end

The prototype opens on Trust Check. Click an alert to jump to the employee's row in the HR module where the data lives, fix or accept it there, and watch the Pay Score reach 100 before Approve unlocks.




Break payroll: you change Aisha's bank account → the Bank Account Check catches it → the live Trust Check drops to 79 and locks Approve → fix it in Employee Records

### How to open it

Double-click **PayShield_Demo.html** in any browser, or run `npm install` and `npm run dev`. 69 automated tests: `npm test`.

### Product pages (click freely)

PayShield: Trust Check, PayShield Agent · HR modules: Employee Records (Personal, Bank), Compliance & Policy Centre (Salary, Compliance policy), Timesheet · Audit: Pay run records

Trust Check only analyses and alerts. Five checks: Employment, Bank Account, Salary, Statutory Compliance and Timesheet. Fixes happen in the HR modules, on the flagged rows, with an on/off switch per check.

### What to show

1. **Break payroll** · Make a real mistake in an HR module → the PayShield check catches it → live Pay Score drops and Approve locks → fix where the data lives
2. **Trust Check** · Pay Score 79, results per HR module, what to fix before payday
3. **Fix where the data lives** · Click an alert, the module opens at that row; fix or accept, the score updates
4. **PayShield Agent** · One prompt → decide each finding (saved to PayShield) → approver → pay run
5. **Pay run records** · Every run, decision and check switch, with who and when

### Suggested 3-minute pitch path

Trust Check (Pay Score 79) → click Daniel's alert → fix in Employee Records › Bank details → PayShield Agent: "check the pay run" → decide each finding A/B/C → Pay Score 100 → **Break payroll**: hand the mouse to someone in the room, let them change a bank account or add S$20,000 → caught live, Pay Score drops → fix it where the data lives → approver → audit record.